

OR Sentinel — Judge Scorecard & Winning Plan

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Summary

OR Sentinel is expected to score about 88 to 92 out of 100 if the four core fixes ship, up from 66 for the audio-only version. That range puts it in finalist territory with a real shot at winning.

Pitch: One ZGX Nano watches 8 operating rooms at once, sees and hears every sponge, and blocks the team from closing if anything is missing. No patient data ever leaves the hospital.

Target user: the circulating nurse, who owns the sponge and instrument count during surgery.

Deadline: Friday, Sept 25 at 8pm. Aim to submit by 7pm.

Scores are an estimate. The hackathon brief has no official rubric, so the criteria below are built from what the brief says it rewards: edge-first inference, a defensible cloud decision, a named user, conditions the cloud cannot meet, metrics, and a working system rather than a model demo.

Scoring rubric

The biggest gains come from the cloud decision (+7) and the Nano and ZRT usage (+6). Those two criteria are where the audio-only version was weakest.

Criterion	Weight	Audio-only (before)	OR Sentinel (target)	What earns the points
Named user and real problem	15	13	14	Circulating nurse, never events, WHO checklist
Why the cloud cannot do it	15	12	13	HIPAA, sub-2-second alert before closing
Explicit, measurable cloud decision	20	9	16	Clear local-vs-cloud rule, patient data stripped, escalation rate measured

Criterion	Weight	Audio-only (before)	OR Sentinel (target)	What earns the points
Uses the ZGX Nano and ZRT well	15	8	14	8 ORs at once, two models on ZRT, scale benchmark
Training on the device	10	6	8	Fine-tuned vision model with a clear base vs. fine-tuned gain
Metrics and evidence	10	6	9	Latency at 1, 4 and 8 ORs, accuracy, memory, cloud data sent
Demo and pitch	10	9	10	Count-mismatch block, chaos mode, command center
Originality	5	3	4	Live and on-premises vs. the existing after-the-fact product
Total	100	66	88	Up to 92 with the stretch features

What will impress judges

Three features carry most of the score: the audio and camera cross-check, the cloud privacy split-screen, and the 8-OR scale benchmark. Never cut these.

Feature	What judges see	Points	Priority
Audio + camera cross-check	Nurse says 10, camera sees 9, alert fires	+8	Must ship
Cloud tier with privacy split-screen	Full patient record on the left, anonymous question sent to cloud on the right	+7	Must ship
8-OR scale benchmark	One Nano runs 8 rooms, latency chart at 1, 4 and 8 ORs	+6	Must ship
Fine-tuned vision model	Base vs. fine-tuned sponge-count accuracy chart	+3	Must ship

Feature	What judges see	Points	Priority
Chaos mode	Pull the cable, kill the vision model, system degrades to audio-only	+2	Should ship
Four-agent council	Counter, Checklist, Critic and Scribe agents shown live	+1	Should ship
Surgical floor command center	8 room tiles in green, yellow and red	+1	Should ship
Retained-sponge risk meter	Risk score rising before a mismatch	+1	Stretch
Tamper-proof log	Hand-edited entry flagged as tampered	+1	Stretch
Voice query	"Sentinel, count?" answered out loud	+0.5	Stretch
Learning flywheel	One real v1 to v2 retrain	+0.5	Stretch

Why HP judges care: the scale benchmark answers "why not a \$500 mini PC?" and reusing HP's Qwen2.5-VL fine-tuning repo shows their full train, push, pull and serve workflow. Put a slide titled "Why this only works on a ZGX Nano."

Scrappy detail judges remember: a photo of your mock OR, built from pharmacy gauze pads and a metal tray, used to create about 150 labeled training images.

Judge Q&A

Weak answers to the first three questions can cost 15 to 20 points in the 3-minute Q&A. Rehearse them word for word.

Likely question	Prepared answer
Why not just use RFID-tagged sponges?	RFID adds a cost to every sponge and needs special readers. We use cameras many ORs already have, and we also cover untagged instruments.
Why does this need a GB10? It runs on a laptop.	A laptop handles one room. One GB10 runs 8 rooms at once, with a 27B reasoning model and a vision model side by side. Here is the latency chart.

Likely question	Prepared answer
Where is the cloud in your edge-to-cloud system?	Anything time-critical or with patient data stays local. Only anonymous, non-urgent protocol questions go to the cloud. We measure the escalation rate and patient data sent, which is zero.
Nurses don't always say counts out loud. Then what?	That is why we added the camera. If audio and vision disagree, or one is missing, the system flags it.
How is this different from the existing OR Black Box product?	That product mostly analyzes recordings after surgery. Ours works live, inside the hospital, and stops the mistake before closing.
What if the AI is wrong?	It is a second set of eyes. The nurse's count stays the official one, and we never silence an alert automatically.
Is the data real?	No. We used acted audio and photos from a mock OR we built. No real patient data was used.

Metrics to show

The brief requires metrics plus how and why they were chosen, so every metric below comes with its reason. The target values are goals to test against, not results yet.

Metric	Target	Why we chose it
Sponge-count accuracy, base vs. fine-tuned	Clear jump, e.g. 60% to 90%	Proves training on the device added value
Count mismatches caught in test scenarios	100%	The core safety claim
p95 alert latency at 1, 4 and 8 ORs	Under 2 seconds at 8 ORs	Justifies the GB10 and the edge-first design
GPU memory used at 8 ORs	Report the real number	Shows the hardware was pushed
Patient data sent to the cloud	0 bytes	Proves the data residency claim

Metric	Target	Why we chose it
Cloud escalation rate	Report the real percentage	Shows the cloud decision is deliberate and measured
Audio vs. vision disagreement rate	Report the real percentage	Shows the cross-check catches real gaps

5-minute demo script

The count-mismatch block at 1:30 is the moment judges will remember, so rehearse it until it is flawless. Record a backup video in case the live demo fails.

Time	Moment	Criterion it scores
0:00	Hook: surgical never events still happen, and a sponge left inside a patient is one of them	Named user and problem
0:30	Command center: 8 OR tiles, OR 6 turns yellow	Nano and ZRT use
1:00	Agent council chats live as the nurse counts sponges	Systems design
1:30	Surgeon says "let's close." Red alert: nurse said 10, camera sees 9. Closing blocked	Demo, cross-check
2:15	Privacy split-screen: full record stays local, anonymous question goes to cloud	Cloud decision
2:45	Chaos mode: pull the cable, kill the vision model, system keeps running audio-only	Resilience
3:30	Evidence: latency at 1, 4 and 8 ORs, base vs. fine-tuned accuracy, 0 bytes sent	Metrics, training
4:15	Mock OR photo and "Why this only works on a ZGX Nano"	HP fit, originality
4:45	Close: "One ZGX Nano per surgical floor. OR Sentinel."	Pitch

Build plan and cut list

Start fine-tuning the vision model Thursday night, because it is the step most likely to break or run long. Everything else can move around it.

When	Person A	Person B	Person C	Person D
Thu evening	Buy gauze and tray, take and label about 150 photos	Serve the 27B model on ZRT, set up faster-whisper	Record 3 acted OR audio clips	Repo, README, start.sh
Thu night	Start fine-tuning with HP's script (runs overnight)	Counter and Checklist agent prompts	Dashboard skeleton with 8 tiles	Benchmark script
Fri morning	zrt pull and zrt serve the fine-tuned model	Fusion engine and Critic agent	Risk meter and alerts	Cloud tier and privacy split-screen
Fri 12pm to 4pm	Base vs. fine-tuned evaluation	Chaos mode and hash chain	Connect everything end to end	Benchmarks at 1, 4 and 8 ORs
Fri 4pm to 8pm	Record the 2-minute video	Final testing	Social posts with required tags	Submit by 7pm

Cut list, in this order if you fall behind:

Cut	Points lost	Expected total after cut
Voice query	0.5	about 91
Learning flywheel retrain (keep as a roadmap slide)	0.5	about 90
Tamper-proof log	1	about 89
Risk meter	1	about 88
Scale 8 ORs down to 4	1 to 2	about 86

Never cut: the audio + camera cross-check, the cloud privacy split-screen, and the scale benchmark. Losing any one of them drops the project back toward the 70s.

Deliverables checklist:

- ☐ Public GitHub repo with code, README and Dockerfile or setup script
- ☐ Metrics with how and why they were chosen
- ☐ 5-minute demo, live or recorded
- ☐ 2-minute public YouTube video
- ☐ Interactive presentation, not static slides

- ☐ SJSU Google Drive: brief, video, links, presentation, diagrams, photos
- ☐ Social posts with all required tags